Signal

The agentic research arm of Stock Sense

For individual investors, students,
and investment clubs

THE TASK

Identify changes worth investigating and inspect the evidence behind them.

THE PROPOSED VALUE

Less repeated research work.
Better visibility into changes.

PS1 explored historical prices. Signal investigates the company behind the stock.

PROPOSED SOLUTION

A question becomes a cited research brief

“Revenue is growing. Is cash flow improving too?”

01

ASK

Choose a company and save a question

02

RETRIEVE

Get filings, figures, and stock price

03

INVESTIGATE

Compare figures and inspect evidence

04

RETURN

Review findings and ask follow-ups

YOUR RESEARCH RECORD

- Verified changes + management explanations + open questions
- Save notes and update the record when a new filing arrives.

APPROACH

Accessible data and a simple build

Source	What it supplies	Refresh
SEC EDGAR	Reports and financial figures	When companies file
Alpha Vantage, planned	Latest end-of-day stock price	Daily, with date shown

PYTHON

Retrieve data and calculate changes

AI AGENT

Select evidence and explain findings

SAVED WORKSPACE

Keep questions, sources, and research history

Planned stack: Python + Streamlit + SQLite + hosted AI model

Resources: API keys, model-call budget, and a host for scheduled checks

A working research prototype

START SMALL

1 company first
Then expand to 5

- One US industry
- Annual + quarterly filings
- Revenue, profit, cash flow, cash, margins, stock price

Risk	Response
Wrong period or units	Validate dates and calculate in code
Unsupported explanation	Show passages and uncertainty
Missing or stale data	Cache sources and show timestamps

IF TIME IS TIGHT

- Keep fewer companies and manual refresh.
- Add daily checks once the core is reliable.

Later: price history, peer comparisons, selected announcements, and news

A clear plan and checks that matter

Milestone	Demonstrable result
First: data pilot	Access works and figures match sources
Midpoint: core prototype	One company, question, and cited brief
Final: tested workflow	Saved research, five-company target, updates

ACCURACY

Check figures and citations on 12–15 research tasks

USEFULNESS

Compare with a summary
Ask users about effort

UPDATES

Replay a historical filing without duplicate reports

Value to test: does Signal reduce repeated work and make important changes clearer?